K_{frak}, Unrolling and the $\tilde{T} \cdot m$ Duality

Why the Fractal Correction Is Not a Property of Time Unrolling

Doc. 333

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Contents

Preliminary note

In the FFGFT corpus, "unrolling" has hitherto always meant **type-I decompactification**: the measurement act selects the mass circle as time, $S^1_m \to \mathbb{R}_t$, discarding the winding number — a lossy step (Doc. 285, Doc. 248, Doc. 270). Doc. 332 introduces alongside this a different operation: the **canonical function pullback** $\Phi = p^* : L^2(S^1_m) \to \mathrm{Per}_{R_m}(\mathbb{R}_t)$, which is lossless and exact — a pure change of coordinates with no loss of information.

This document places both operations side by side, clarifies the role of K_{frak} in each case, and shows that the recursion follows necessarily from the $\tilde{T} \cdot m$ duality.

.1 Two Meanings of "Unrolling" in the Corpus

Type I: the measurement act (lossy)

The meaning used so far in the corpus (Doc. 285 § "How FFGFT unrolls"): the embedded observer selects the mass circle S^1_m as the time axis. The map

$$S^1_m \longrightarrow \mathbb{R}_t, \qquad \tilde{T} \mapsto t,$$

is **lossy**: the winding number is discarded, the discrete spectrum of the compact circle does not appear in $\mathbb{R}_t$. This is not a mathematical operation of the theory but the bridge to observation — the torus remains the fundamental structure, $\mathbb{R}_t$ is the image (Doc. 285). At the same time the recursion ξ^n unrolls into a continuous dilation flow $t \mapsto \xi t$ with log-period $|\ln \xi| = \ln 7500 \approx 8.92$ (Doc. 285 § "The recursion unrolls along with it").

Type III: the coordinate change (lossless)

The meaning introduced in Doc. 332: the canonical function pullback

$$\Phi = p^* \;:\; L^2(S^1_m) \;\longrightarrow\; \mathrm{Per}_{R_m}(\mathbb{R}_t), \qquad (\Phi f)(t) = f(t \bmod R_m),$$

is an **exact isometry** — no loss of information, no winding number discarded, Fourier basis and Parseval identity preserved. It is a change of representation within isomorphic Hilbert spaces (R96, Doc. 332). The lossy direction lies at $\mathbb{R}_t \to S^1_m$, not at the pullback itself.

	Type I (measurement act)	**Type III (pullback)**
Operation	$S^1_m \to \mathbb{R}_t$	$L^2(S^1_m) \cong \mathrm{Per}_{R_m}(\mathbb{R}_t)$
Loss	winding number discarded	none
Character	physical act	mathematical change of representation
Winding number	not preserved	preserved
Information loss	yes	no
Corpus references	Doc. 285, 248, 270	Doc. 332 (R96)

.2 The Role of K_{frak} in Both Cases

At the type-III pullback: invisible

A Laplace operator on the compact T^4 is local — it cannot carry a cumulative quantity (Doc. 314). K_{frak} therefore does not enter through the operator but through the **bridge constants** E_0, v at the scale conversion (Doc. 314, R72).

In the rolled-up state on S^1_m the period $R_m = \hbar/(mc^2)$ is exact in natural units — no K_{frak}. Only when m is substituted from the mass formula does R_m carry the correction:

$$m_i = r_i \cdot \xi_0^{p_i} \cdot v \cdot K_{\text{frak}} \quad \Rightarrow \quad R_m^{\text{corr.}} = \frac{\hbar}{m_i^{\text{bare}} \cdot K_{\text{frak}} \cdot c^2} = \frac{R_m^{\text{bare}}}{K_{\text{frak}}}.$$

K_{frak} sits in the *choice of* R_m, not in the pullback mechanism. The pullback itself is exact in both cases (bare or corrected).

At the type-I measurement act: visible as a scale correction

At the lossy unrolling, K_{frak} appears as a correction factor of the bridge constants: the ratio of the two E_0 values (bare and α-anchored) is $1/\sqrt{K_{\text{frak}}}$ (R72, Doc. 314). The correction belongs to the *scale structure* — the transition between the compact and the observable representation — not to the time geometry itself.

	Type I	**Type III**
K_{frak}	visible in bridge constants (E_0 ratio, mass formula)	invisible at the isometry step; sits in R_m via m
Where exactly	at the SI transition	in the choice of period

.3 Mass and Time Unroll Together

In the FFGFT compactification, $\tilde{T}$ and m are a **single geometric dimension**:

$$\tilde{T} \cdot m = 1 \qquad \text{(fundamental relation, Doc. 285).}$$

They are not two independent quantities. At the type-III pullback, the pullback therefore unrolls m and t *together*: the entire mode $f \in L^2(S^1_m)$ is unrolled and the ratio $\tilde{T}/m$ remains constant. The physics does not change, only the coordinate representation.

Unrolling t alone — dropping R_m — would violate $\tilde{T} \cdot m = 1$ and is structurally inadmissible. K_{frak} is *not* the price of unrolling, but the price of the **artificial separation** of m and t at the transition to absolute SI units.

.4 The Recursion Is Necessarily Bound to the Duality

Stable particle: Case B

For a stable, isolated particle ξ is frozen (Case B, Doc. 315, Doc. 313): the winding number is rational $74/75$, closure occurs after exactly 75 revolutions. $K_{\text{frak}} = 74/75$ is the closed bookkeeping of this rolled-up geometry — exact in fractional arithmetic, because ξ is set rationally (Doc. 315).

Dynamics: Case C and the recursion chain

As soon as dynamics enter — interaction, RG flow, mass change — a deeper consequence of $\tilde{T} \cdot m = 1$ comes into play: if m changes, $\tilde{T} = 1/m$ changes necessarily with it. The period $R_m = \hbar/(mc^2)$ changes, the time scale of the physics changes, and this feeds back onto the effective mass. The recursion (Doc. 295, Doc. 146) reads:

$$\xi_{n+1} = \xi_n \, (1 - 100\, \xi_n).$$

The closure deficit per revolution no longer remains constant but runs cumulatively. The rational 75-closure becomes a logarithmic spiral (Case C, Doc. 295):

$$\rho = 75\, e^{\beta/2\pi} \qquad \text{(equiangular spiral, self-similarity ratio } e\text{).}$$

In Case C, K_{frak} is no longer a constant but an effectively scale-dependent quantity. The multiplicative form $(1-\xi)^{100}$ is the natural bookkeeping of this running case; its distance from $74/75$ — the binomial term $4950\,\xi^2$ (Doc. 315) — is precisely the difference between frozen and running recursion.

Why Case A is excluded

Case A ($K_{\text{frak}} = 1$, closure after one revolution) would mean: $\tilde{T} \cdot m = 1$ generates no recursion, because m never changes — a static, interaction-free universe with $D_f = 3$ exactly. This contradicts the observed particle structure and the basic framework.

Summary of the three cases

Case	ξ	K_{frak}	**Binding to duality**
A	constant, $d = 0$	$= 1$	no feedback, excluded
B	frozen	$= 74/75$ exact	stable mode, duality at rest
C	running (recursion)	scale-dependent	duality generates running feedback $m \leftrightarrow \tilde{T}$

$K_{\mathsf{frak}} = 74/75$ is the **projection of Case C onto Case B**: exact in the rolled-up, static description; first approximation in the dynamic one.

.5 Mode Specificity, Non-Closure and Superposition

Each mode carries its own period

The period $R_m = \hbar/(mc^2)$ is mode-specific: every particle i with torus mode $\mathcal{T}_i = (n_i, l_i, j_i)$ carries its own period. Unrolling (Type III or Type I) is therefore always particle-specific — it unrolls one particular mode, not a mixture. Simultaneously unrolling different periods R_{m_e} and R_{m_μ} would be structurally inadmissible, because $\tilde{T} \cdot m = 1$ would be violated for both at once.

The non-closure condition (Doc. 193)

The same mode specificity underlies the non-closure condition (Doc. 193): the prefactors r_i are not numbers in a field but labels of discrete geometric spaces — fully determined by $\mathcal{T}_i$. **Multiplicative** combination of different modes,

$$(r_i \cdot r_j,\ p_i + p_j) \notin \mathcal{P} \qquad \text{for } i \neq j,$$

yields no FFGFT particle — the product lies outside the set of admissible pairs **[B]**. K_{frak} sits in the mode-specific choice of R_m; a cross-mode K_{frak} would be the same inadmissible cross-product.

Why superposition is nevertheless permitted — and where the difference lies

In quantum mechanics, superpositions of different states are not only permitted but fundamental. An electron in the sp^3 hybrid orbital of a carbon atom is a linear combination of s and p states; an excited atom is in a superposition of ground state and excited state. The operation is **additive**:

$$|\psi\rangle = c_1\,|\psi_1\rangle + c_2\,|\psi_2\rangle\,, \quad |c_1|^2 + |c_2|^2 = 1.$$

This produces no new (r, p)-pair, no new mass, no new torus mode — and therefore does not contradict the non-closure condition.

The crucial difference between QM and FFGFT: In standard QM, time t is not a quantum-mechanical operator and not part of the Hilbert space — it is simply *absent* as a physical quantity of the system. The Schrödinger equation treats t as a classical external parameter that evolves the state from outside, without itself belonging to the system. A superposition

$c_1 |\psi_1\rangle + c_2 |\psi_2\rangle$ therefore carries no mode-bound time structure — t does not belong to the superposition but stands outside it.

In FFGFT, time is geometrically bound to mass: $\tilde{T}_i = 1/m_i$. Two modes with different masses $m_1 \neq m_2$ carry different time scales $\tilde{T}_1 \neq \tilde{T}_2$. A superposition of such modes would not merely mix amplitudes but overlay incompatible time scales — structurally more problematic than in QM, where this conflict does not arise.

In the FFGFT corpus, superposition is therefore described as **intermediate torsion within a mode** (Doc. 174): a metastable intermediate state between stable torsion extremes of the *same* mode, not a cross-mode superposition. Decoherence drives this intermediate state back into one of the stable configurations.

	Standard QM	**FFGFT**
Time	absent: no operator, no Hilbert space element	$\tilde{T} = 1/m$, mode-bound, geometric
Superposition	additive, t stands outside	intermediate torsion within one mode
Cross-mode	algebraically permitted	incompatible $\tilde{T}$, structurally problematic
Non-closure	not relevant	$r_i \cdot r_j \notin \mathscr{P}$, multiplicatively forbidden **[B]**

The consistent boundary

The linearity of quantum mechanics — the superposition principle — and the non-closure condition protect the same boundary from two sides: atoms and bound states generate no new fundamental particles, no new torus modes, no new (r, p)-pair. The connection between FFGFT and established atomic physics runs through the derived quantities α and m_e (Doc. 011, 044, 006) — quantum mechanics takes these as input and derives binding energies and spectra from them, without touching the mode structure.

> *"Unrolling" in the FFGFT corpus has hitherto always meant the lossy measurement act (Type I, Doc. 285): winding number discarded, $S^1_m \to \mathbb{R}_t$. The function pullback from Doc. 332 is a different, lossless operation (Type III): an exact change of representation that unrolls m and t together and at which K_{frak} plays no role. K_{frak} sits in the choice of the period R_m via the mass formula and becomes visible only at the transition to absolute SI values. The recursion $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ is necessarily bound to $\tilde{T} \cdot m = 1$: every mass change entails a time change, which feeds back recursively onto the mass. Case B (stable particle, $K_{frak} = 74/75$) is the approximation for frozen duality; Case C (running recursion, log-spiral) is the complete dynamic description."*

References: Doc. 133 (K_{frak} derivation), Doc. 146 (recursion), Doc. 248 (observer), Doc. 270 (type-I theorem), Doc. 285 (decompactification), Doc. 295 (log-spiral), Doc. 313 (three cases), Doc. 314 (local operator, bridge constants), Doc. 315 (form question K_{frak}), Doc. 332 (pullback isometry, R96).